

TrES-3b

R-2733

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_110621 · created 2026-09-02T00:33:58+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2455733.74142 +/- 0.0009 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.159 +/- 0.016
Transit depth (Rp/Rs)^2	2.52 +/- 0.5 [%]
Orbital Inclination (inc)	83.02 +/- 0.7
Airmass coefficient 1 (a1)	1.251 +/- 0.014
Airmass coefficient 2 (a2)	-0.211 +/- 0.019
Scatter in the residuals of the lightcurve fit is	1.09 %
Best Comparison Star	None
Optimal Aperture	3.54
Optimal Annulus	8.03
Transit Duration (day)	0.0636 +/- 0.0052

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.159 ± 0.016 vs 0.16309 (deviation 0.26σ)
Duration fit vs published	0.0636 d vs 0.0567 d (deviation 1.33σ)
Mid-time O–C	-1.3 min
Depth SNR	9.9
Residual scatter	1.09 %

Light curve

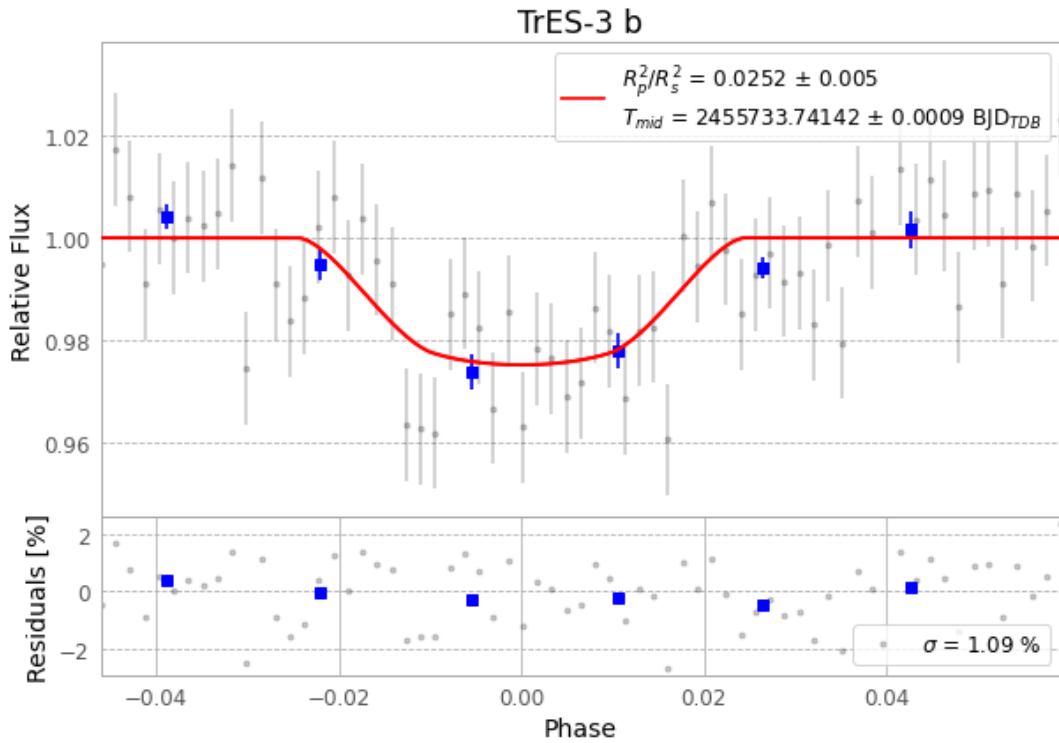


Figure 1 — FinalLightCurve_TrES-3 b_2011-06-20.png (SHA-256 c24296d8bbfc98a0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [293, 207], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1dce2cbb603f8ac3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

67 FITS frames (44 MB), TRES-3110621041513.FITS → TRES-3110621073313.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-110621.log (ddb733a82c6...), FinalLightCurve_TrES-3 b_2011-06-20.png (c24296d8bbfc...), FinalLightCurve_TrES-3 b_2011-06-20.csv (1dc1e1479611...)

Compute resources

Wall time 141.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-02 00:33Z.